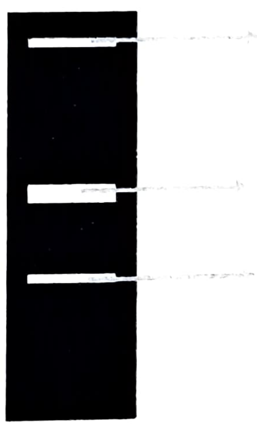
Examination	:	End-Semester Examination – May/June 2026
Name of the Course	:	B.Tech. (IT & Mathematical Innovations)
Name of the Paper	:	Genomics and Proteomics (GE)
Unique Paper Code	:	3124000022
Semester	:	VI
Duration	:	2 hours
Maximum Marks	:	60

Instruction to students:

*Attempt **any four** questions from the following. All questions carry equal marks. Draw schemes, labelled diagrams wherever necessary.*

1. Describe how the contamination of genomic DNA is avoided during plasmid DNA isolation.
   A researcher, upon separation of plasmids on agarose gel, got three bands as below. Identify the bands.



2. Describe in detail the PCR cycle. Give a schematic presentation of reverse transcriptase PCR. If PCR amplified for 12 cycles, a single DNA molecule would produce how many DNA molecules?

3. With the help of schematic presentations, differentiate between a cloning and an expression vector. What is the use of antibiotic resistance-based screening?

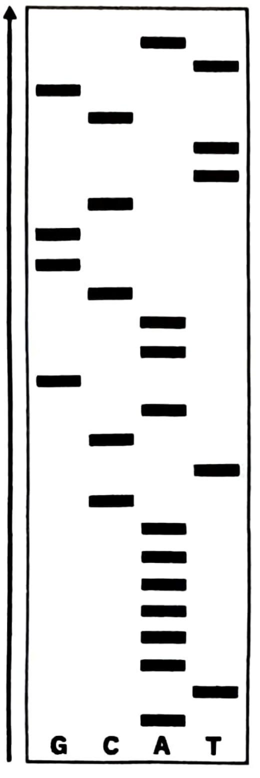
4. How is protein sequencing done using mass spectrometry? Write the advantages and disadvantages of MALDI-based sequencing over N-terminal sequencing.

5. Give a detailed classification of Restriction Enzymes. Differentiate between frequent cutters, rare cutters, sticky end cutters and blunt end cutter REs.

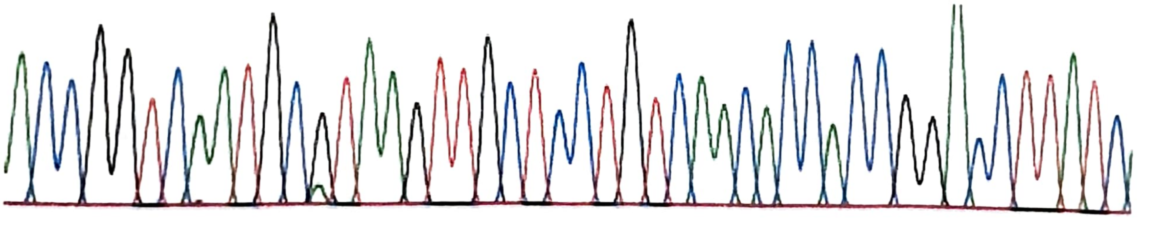
[PTO]

6. Given below are two sequencing data. Identify and write the DNA sequences in "A" and "B".

A



B



Here green, red, blue and black indicate A, T C & G respectively.

7. Write short notes on the following:
   A. CRISPER-CAS9
   B. Protein Chip
   C. siRNA

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